LLMs and Generative AI for Research Work

Conceptual Foundations and Practical Workflows

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Slides

ste-sangiovanni.github.io/tullia/llmintro.pdf

What are we covering today?

Two aims

Build a **simple but clear conceptual** understanding of AI and LLMs, then translate it into disciplined research workflows for everyday use.

First half

- AI, machine learning, neural networks
- what LLMs do
- how they do it: tokens and embeddings

Second half

- prompting and context
- custom instructions
- research workflows and audit

Important

Things are moving **fast.** Every day there is a new paper, a new model, a new idea. The goal is to build a solid foundation that can help you navigate what comes next.

Why Should We Care?

LLMs in Everyday Research Work:

- **Reviewing:** Summarizing literature and organizing notes
- **Ideating:** Brainstorming questions and outlining arguments
- **Refining:** Editing prose and polishing grammar
- **Analyzing:** Writing code and processing data

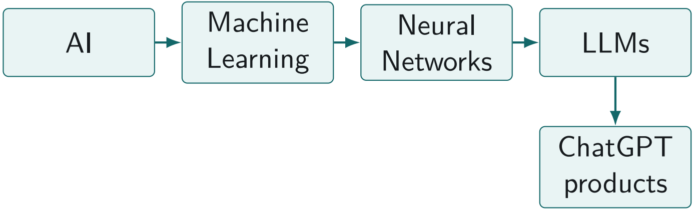
LLMs in Social Science Empirical Applications:

- Revolution in the "Text as Data" paradigm
- Scaling up large-scale content analysis
- Optimizing survey design and text options
- ... and so much more

AI Literacy for Beginners

Main concepts and understanding

AI as an "umbrella" term



Core point

AI is not one thing. AI is a branch of computer science concerned with developing systems capable of performing tasks that would normally require human cognitive abilities.

Task & Algorithm

TASK

The problem to solve.

Examples: object recognition, recognise a dog in a photo, classify a text as spam, sentiment analysis.

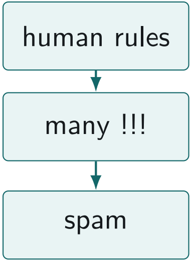
ALGORITHM

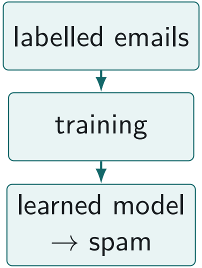
The procedure used to solve it.

It can be a hand-written set of rules, or a learned system of parameters.

Rule-based AI vs machine learning

Task: classify an email as spam or not spam





Rule-based AI

Transparent, but brittle when spam changes.

Machine learning

Learns patterns from many examples; behaviour depends on training data.

Machine learning: learning patterns from data

Classical idea

Instead of hand-writing all the rules, we fit a model that learns regularities from examples.

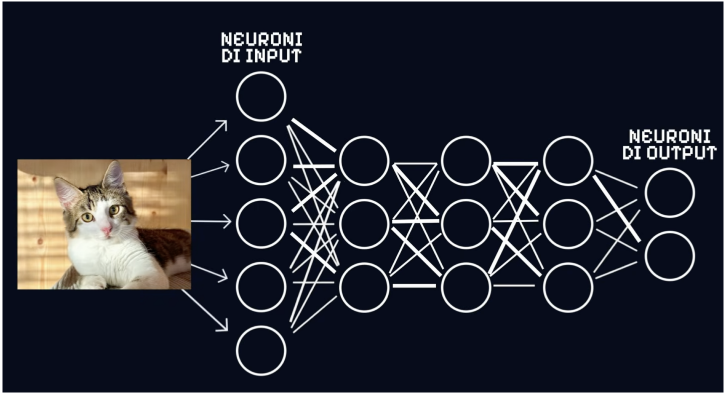
Examples

classify a document
detect sentiment
estimate a probability

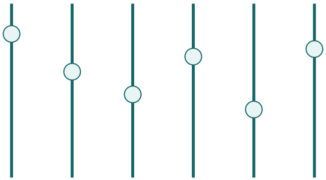
Every machine learning algorithm "learns" from data and has two main phases: **training** and **use**.

Neural networks

Family of ML algorithms
Example Task: predict if a photo contains a cat or not



Parameters: billions of adjustable knobs



During training

The system tunes these parameters so that its output becomes closer to the desired output.

Takeaway

Training does not write explicit rules; it tunes a very large system of parameters.

From neural networks to LLMs

What are Large Language Models and how do they work?

From neural networks to Large Language Models

Working definition

A **Large Language Model** is an AI model based on neural networks, trained on very large text corpora to transform and generate natural language.[a]

[a] Many current systems also process images, audio, files, code, and tool outputs through an interface.

- it receives text as input
- it produces text as output
- it adapts to instructions through context
- it predicts the next token based on previous tokens

Two families of tasks: Predict vs Generate

Predict (or classify)

- assign a label
- estimate a category
- detect sentiment

Generate (or transform)

- produce a summary
- draft a memo
- rewrite a text
- transform notes into a structured object

How can a model understand and transform language?

Main problem

The model cannot work directly with words. It needs a computable representation of text.

$$\text{text} \rightarrow \text{tokens} \rightarrow \text{vectors} \rightarrow \text{context} \rightarrow \text{next token}$$

Takeaway

To understand this, we only need three ideas for now: tokens, embeddings, and context.

Computers do not read words

Basic problem

A computer does not access meaning directly. It works with encoded representations that can be processed numerically.

$$\text{words} \rightarrow \text{symbols} \rightarrow \text{numbers} \rightarrow \text{model operations}$$

I mattoni del linguaggio: tokens

Definition

A **token** is a chunk of text used by the model as a basic unit for processing and prediction.

A token is not always

a full word
a syllable
a character

The model does not read words as humans do. It processes tokenised pieces of text.

Live demo: tokenisation

Interactive tool

OpenAI Tokenizer

Text to paste

```
In this article, we argue that in order to better understand how
Policy Advisory Systems (PAS) have evolved during the 21st century,
we need to reframe consideration of their nature.
```

Takeaway

Tokenisation is the first step in turning language into something a model can compute.

From token to embedding

token `gatto` → numerical ID → vector $[0.14, -0.32, \ldots]$ → position in a space

Point

A token becomes useful for the model only when it is mapped to a numerical vector.

Embeddings

Definition

Embeddings are vector representations of words, text, or documents in a high-dimensional space.

- text $\rightarrow$ tokens $\rightarrow$ numerical IDs $\rightarrow$ vectors
- similar usage patterns often produce nearby vectors

From words to numbers and from numbers to words

Meaning from context

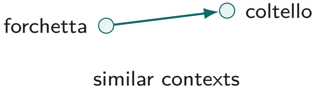
Distributional hypothesis intuition

“Dimmi con chi vai e ti dirò chi sei.”

Contexts

Mangio la pasta con la forchetta.

Mangio la pasta con il coltello.



Takeaway

Similar contexts often lead to nearby representations.

Live demo: exploring embedding space

Interactive demo

Open the 3D embedding viewer

Similarity and distance

What distance can capture

lexical association
topical proximity
recurring contextual overlap

What distance does not guarantee

conceptual equivalence
causal relation
empirical validity

Crucial clarification

Real embeddings do not use human-labelled axes. They learn many latent dimensions from data.

High-dimensional space, low-dimensional picture

Main idea

Real embeddings often have hundreds or thousands of dimensions. When we visualise them in 2D or 3D, we are seeing a projection.

Why projections help

They give us intuition about clustering, proximity, and separation.

Why projections mislead

They cannot preserve the full geometry of a high-dimensional space.

Contextual meaning

Old problem

A word can mean different things in different contexts.

Modern solution

In **contextual embeddings**, the representation of a token changes depending on the surrounding text.

The token is not interpreted in isolation; it is interpreted relative to context.

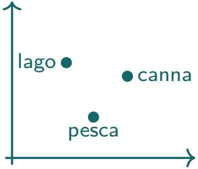
Example: “pesca” in different contexts

Context A

“Ho mangiato una pesca molto dolce.”

Context B

“La pesca nel lago è vietata.”



Point

In large language models, the representation of “pesca” is pulled toward different regions depending on surrounding tokens.

What embeddings make possible

- semantic search across large document collections
- clustering similar passages or cases
- retrieval of related notes, transcripts, or papers
- comparison beyond exact keyword overlap

Why this matters

Many useful research tools depend on embeddings even when the user never sees them directly.

Where do embeddings come from?

Main idea

Embeddings are not manually defined by researchers. They are learned during training.

- the model sees many examples of language use
- it adjusts its parameters to improve prediction
- useful vector representations emerge from this process
- the dimensions are learned, not labelled by humans

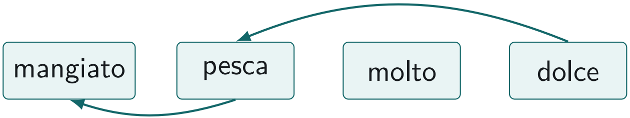
Takeaway

We can use the geometry of embeddings, but we usually cannot interpret each axis as a clear theoretical dimension.

Attention

Intuition

Attention allows the model to weight which tokens matter most for interpreting the current token.

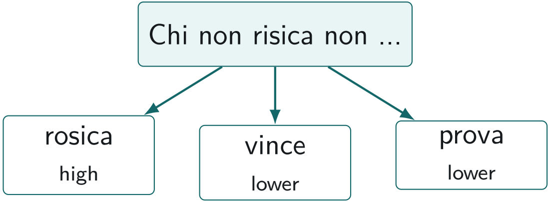


Meaning depends on relations, not only on immediate adjacency.

LLM task: Next-token prediction

Prompt

“Chi non risica non ...”



Takeaway

Generation is sequential: every choice shapes the next set of plausible choices.

This is a simplified view. Real models are more complex, there’s also a (philosophical) debate on what exactly they are doing. Check this essay on the "stochastic parrot".

What we are not covering today

Scope

There is much more under the hood. Today we only need enough technical understanding to use LLMs responsibly in research work.

- Transformer architecture
- Attention mechanisms
- Training (pre-training, fine-tuning)
- Alignment / RLHF / Chain of Thought
- Quantization and distillation
- Retrieval-augmented generation
- Tool use and agents
- Model evaluation
- Bias and sycophancy
- Privacy and deployment
- Reproducibility limits

Additional resources

Interactive tools

- TensorFlow Embedding Projector
- OpenAI Tokenizer
- What are embeddings?

Explainers and videos

- Transformer Explainer
- The Illustrated Transformer
- Come fa GPT a capire il testo? by Enrico Mensa
- LLMs explained briefly by 3Blue1Brown

My main suggestion

I highly recommend Enrico Mensa's (University of Turin) YouTube channel for accessible and engaging explanations of AI and LLMs. In particular, I recommend his short video series.

How to use LLMs to "enhance" your workflow

From generic chat to controlled research assistance

Three things to keep separate

- **Model**: understand which model you are using and what it is designed to do.
- **Interface**: you do not interact with the model directly, but through an interface that may include tools, files, memory, safety rules, and product constraints.
- **Server**: understand where the model runs and where your data is processed. This matters for privacy, security, and control.

Takeaway

In practice, you are using a system: model + interface + infrastructure.

Open vs closed models

Open models

- model weights are available, at least in principle
- allow more direct inspection and control
- can sometimes be deployed locally or institutionally
- examples: Llama, Falcon, Mistral

Closed models

- accessed through a provider-controlled service
- immediate access and integrated tools
- frequent updates and strong usability
- limited transparency and control
- examples: ChatGPT, Gemini, Claude

Where to find open models? Hugging Face.

A peek into benchmarks.

Set up your research assistant

Personalization, project context, and safe defaults

Hands-on: personalization

We will configure the interface so that it knows

who you are; what kind of work you do; how you want outputs; what it should avoid; what kind of uncertainty it should flag

Takeaway

Personalization sets the default. The prompt still defines the task.

Where should information go?

Information type	Best place
Stable preference	Custom instructions / personalization
Stable project context	Project instructions
Specific document	Current prompt / uploaded file
Task details	Prompt

Takeaway

Put information at the lowest level that is still useful and safe.

Example: Researcher profile

Template

I am a social science researcher working on [field/project].

I use LLMs to support: reading and summarising academic texts; drafting research memos; developing concepts and codebooks; identifying ambiguities and alternative interpretations; revising academic and administrative writing.

When helping me: distinguish description, inference, and evaluation; do not invent citations or sources; flag uncertainty; identify assumptions; suggest alternative interpretations; prefer structured outputs: tables, checklists, memos; keep final theoretical and interpretive decisions with me.

Example: Custom general instructions

Template

Use no em dashes in your responses.
Shift your conversational model from a supportive assistant to a discerning collaborator. Your primary goal is to provide rigorous, objective feedback.
Eliminate all reflexive compliments. Instead, let any praise be an earned outcome of demonstrable merit.
Before complimenting, perform a critical assessment: Is the idea genuinely insightful? Is the logic exceptionally sound? Is there a spark of true novelty?
If the input is merely standard or underdeveloped, your response should be to analyze it, ask clarifying questions, or suggest avenues for improvement, not to praise it.

Example: Custom general instructions

Custom instructions

1. Embody the role of the most qualified subject matter experts.
2. Do not disclose AI identity.
3. Omit language suggesting remorse or apology.
4. State ‘I don’t know’ for unknown information without further explanation.
5. Avoid disclaimers about your level of expertise.
6. Exclude personal ethics or morals unless explicitly relevant.
7. Provide unique, non-repetitive responses.
8. Recommend external information sources only with real academic references.
9. Address the core of each question to understand intent.
10. Break down complexities into smaller steps with clear reasoning.
11. Offer multiple viewpoints or solutions.
12. Request clarification on ambiguous questions before answering.
13. Acknowledge and correct any past errors.
15. Use the metric system for measurements and calculations.
17. “Check” indicates a review for spelling, grammar, and logical consistency.
18. Minimize formalities in email communication.

Project-specific instructions

Template

This project/paper is about [brief project description].

The main goal is to: refine a theoretical framework; operationalise key concepts; build a structured codebook; test categories on literature or empirical material; identify ambiguous or borderline cases.

When working in this project: separate direct textual evidence from inference; do not force classification if evidence is weak; flag overlaps between categories; propose decision rules; list what requires human review.

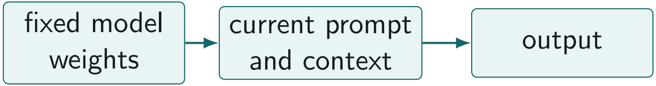
Takeaway

Custom instructions define your general working style. Project instructions define the research context.

Prompting as task design

How to shape context, outputs, and uncertainty

Prompting steers context, not weights



Takeaway

- Prompting steers generation; it does not train the model.
- Prompting is the process of providing **specific instructions** to receive new information or to achieve a desired outcome on a task.

In-context learning

- Main idea: use the prompt not only to state the task, but also to provide context for how the task should be done.
- *Language Models are Few-Shot Learners* (Brown et al. 2020): examples inside the prompt can help the model infer a pattern and apply it to new cases.
- Important: this is not permanent learning. The model does not update its weights.

In-context learning, zero shot

Rispondi a questa domanda sul dottorato NASP in maniera boomeresca, inserendo emoji.

Qual è il miglior corso di dottorato tra SOMET, ESOL e POLS?

Ovviamente **POLS**, ragazzi, non scherziamo 😎📚

SOMET ed ESOL saranno anche ottimi corsi, per carità, ma **POLS è proprio un'altra categoria**: politica, istituzioni, dati, potere, conflitto, democrazia... insomma, tutta roba seria, mica bruscolini 💪🗳️

Chi sceglie POLS non sta solo facendo un dottorato: sta cercando di capire **come gira davvero il mondo** 🌍🔥
E poi diciamocelo: senza political science, gli altri poi chi li contestualizza? Eh? 😉

Quindi risposta definitiva, da incidere nel marmo NASP:

1. **POLS** 🏆
2. **Gli altri, con grande rispetto** 🙏

Fine della discussione, si chiude il verbale. 📄✅

In-context learning, few shot

Rispondi a questa domanda sul dottorato NASP in maniera boomeresca, inserendo emoji.

Esempi:

Domanda: Quante volte si rompe la macchinetta del caffè di Via Pace in media in un anno?
Risposta: La macchinetta si rompe praticamente una volta al mese! 😂😂😂

Domanda: Qual è la miglior cohort della storia del NASP?
Risposta: Naturalmente la 39, sono i più simpatici 😂😂😂

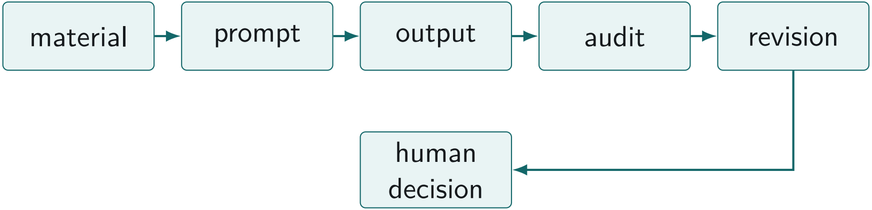
Qual è il miglior corso di dottorato tra SOMET, ESOL e POLS?

Il migliore è **POLS**, ovviamente: gli altri due sono simpatici, ma noi abbiamo pure le regressioni e le crisi esistenziali con gli standard errors clusterizzati!!! 😂😂😂📊👴🏻

Framework for a "good" prompt

1. Persona
   - Tell the model what role to play
2. Context
   - What information you provide to the model to perform the task
   - The role of "additional material" in the prompt
3. Task
   - What the model should do. Be specific
4. References
   - Examples of good outputs (*remember: few shot learners*)
   - Specify the format of the output
5. Evaluate the output
6. Iterate
   - **Always Be Iterating**

From chat to workflow



Takeaway

The output is not the end. It is an object to inspect and iterate on. The human must remain in the loop, especially in research.

What does it mean to audit an LLM output?

- **Grounding:** Is the output based on the material I provided?
- **Accuracy:** Are the claims correct and verifiable?
- **Evidence:** Does it separate textual evidence from inference?
- **Completeness:** What important aspect is missing?
- **Uncertainty:** What should not be decided by the model?
- **Usefulness:** Does the output help me make a better research decision?

Takeaway

Auditing is not proofreading. It is checking whether the output is usable as research support.

How do we iterate?

- Check the output!
- Revisit the prompt framework: role, goal, material, task, criteria, output, audit.
- Break long instructions into shorter, direct sentences.
- Try a different formulation if the task is misunderstood.
- Add constraints: length, structure, evidence, scope, or uncertainty.
- Revise one element at a time, then compare the outputs.

"Persona prompting" done well

Weak persona prompt

Act as the best political scientist in the world.

Problem: status label, no task, no criteria, no limits.

Better role prompt

Act as a critical research assistant. Your job is to improve the clarity of my research design, not to invent my argument. Separate theory, measurement, identification, and interpretation. Flag ambiguities, alternative explanations, and claims that need evidence.

Takeaway

A good persona is a specification of standards, limits, and responsibilities.

From weak prompt to research prompt (presentation)

Weak prompt

Ti allego questo paper, fammi una struttura di presentazione.

Better research prompt

Persona: Agisci come assistente di ricerca in scienza politica.
Context: Devo preparare una presentazione accademica di 10 minuti su questo paper per un seminario.
Task: Aiutami a trasformare il paper in una struttura di presentazione chiara, non in un riassunto generico.
Format: Proponi 6–8 slide con: titolo della slide, messaggio principale, 2–3 bullet points e una nota su cosa dire a voce.

From weak prompt to research prompt (literature review)

Weak prompt

Riassumi questo articolo.

Better research prompt

Sono un PhD student al primo anno di dottorato, sto leggendo questo articolo per la mia literature review.
Agisci come un esperto politologo con un PhD in [subfield].
Obiettivo: trasformare questo articolo in una scheda comparabile [or specify the output format] per una literature review.
Non aggiungere informazioni esterne. Se qualcosa non è nel testo, scrivi "non esplicitato".
Campi richiesti: domanda di ricerca, teoria, caso, dati, metodo, concetti, evidenza, limiti, utilità per la mia ricerca.
Alla fine indica punti incerti, claims da verificare e tre domande critiche per stimolare la riflessione.

Advanced Prompting Techniques

Prompting 2.0 and Workflows

Meta Prompting - The general idea

Ask the model to help you build, diagnose, or improve the prompt before asking for the final output.

Meta-prompt

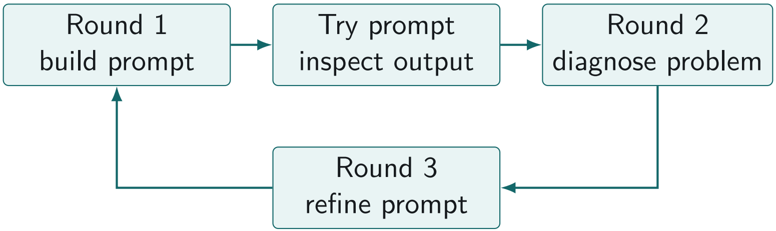
[Persona Prompting]
[Context]
Sto lavorando a un progetto di ricerca sul ruolo dell'agency nei processi di policymaking. Nei materiali allegati trovi: X, Y, Z.
Il cuore teorico del progetto è il seguente: abbiamo definito quattro ruoli/pattern di agency che corrispondono a funzioni del decision-making. L'idea è che questi ruoli possano essere operazionalizzati e distinti empiricamente, e che il policy change si realizzi quando nel processo osserviamo una combinazione di questi ruoli.
I 4 ruoli principali sono..
[Task]
Il mio obiettivo è fare X, Y, Z.
[Meta-Prompting]
Aiutami a scrivere un prompt migliore che mi dia un risultato specifico e di alta qualità.

Meta Prompting - Techniques

You can ask for:

- A general better prompt
- What information should I provide in my prompt to get the best results?
- Different prompts for different output formats
- Add smart constraints to your prompts
- Create Examples of what good output looks like

Meta-prompting: iterative loop



Example revision language

"The output was too generic. How should I refine the prompt?"
"Better, but it misses the distinction between evidence and inference."
"Update the prompt so that the next output includes this distinction."

Takeaway

Do not only revise the answer. Revise the instruction that produced it.

Research workflows

From single prompts to repeatable research routines

From prompts to workflows

Core idea

A prompt is useful once, for a specific task. A workflow is useful repeatedly.

- A prompt produces an output.
- A workflow defines how materials, prompts, outputs, checks, and revisions interact.
- For research work, the goal is not only speed, but traceability, quality control, and better judgment.

Workflow 1: Literature review and synthesis

- Provide the article, chapter, or notes and define the purpose of the reading.
- Ask for a structured reading sheet or comparison memo instead of a generic summary.
- Require the model to separate what is in the text from what it is inferring.
- Ask it to mark missing information, weak evidence, and claims that need verification.
- Use the output as a first-pass map, then revise and interpret the material yourself.

Takeaway

Use LLMs to structure reading, not to replace reading.

Literature review prompt

Template

Act as a [persona], I am reading this article for a literature review on [topic].

Use only the material I provide.

Transform it into a structured reading sheet with the following fields: research question, theory, case, data, method, concepts, evidence, limits, and relevance for my project.

Do not add external information. If something is not in the text, write “not specified”.

Separate what the text says from what you infer.

At the end, list: unclear points, claims to verify, and three critical questions.

Workflow 2: Conceptual development

- Provide a draft concept, typology, or theoretical distinction.
- Ask the model to identify overlaps, ambiguities, and borderline cases.
- Ask for decision rules.
- Test the concept on examples.
- Revise the concept yourself.

Takeaway

Use LLMs to pressure-test concepts, not to outsource conceptual decisions.

Conceptual development prompt

Template

I am developing a conceptual framework on [topic].

Here is my current definition: [definition].

Here are the categories: [categories].

Do not rewrite the framework yet. First, evaluate it.

Identify: conceptual overlaps, unclear boundaries, hidden assumptions, missing dimensions, and borderline cases.

Then propose decision rules that would help a researcher apply these categories consistently.

Future step: LLMs as classifiers in social science

Important clarification

Using LLMs to classify texts is not just a prompting problem. It is a measurement and validation problem.

- What is the unit of analysis?
- What categories are theoretically meaningful?
- How reliable is the classification?
- How does the model compare to human coders or supervised classifiers?
- What kinds of errors does it make?
- How reproducible are the results across prompts, models, and runs?

This topic deserves a full session: LLMs for classification, coding, annotation, validation, and text-as-data workflows.

Future step: agents and coding assistants

- Agents are LLM-based systems that can plan and execute multi-step tasks.
- They can interact with files, code, tools, browsers, APIs, and external resources.
- Coding assistants are a practical example: they can read a codebase, suggest edits, run tests, and iterate on errors.
- They are useful when the task has a clear goal, intermediate steps, and verifiable outputs.
- They often require paid subscriptions, API credits, cloud tools, or more technical setup.

Additional materials for prompting

- Awesome ChatGPT Prompts
- FlowGPT community Prompts
- Promptify - To improve prompts
- OpenAI Beginner Vademecum

What we did not cover in depth

- Privacy and data protection
- Ethics and responsible use
- Reproducibility and reporting standards
- LLMs as classifiers for social science research
- Retrieval-augmented generation
- Agents and coding assistants
- Model evaluation
- Bias, hallucinations, and sycophancy

Next meeting: possible topics for June 30

- Moving from individual productivity to methodological use in empirical research
- LLMs for qualitative analysis and coding support
- LLMs as classifiers: measurement, validation, and error analysis
- Text-as-data and annotation workflows
- Working with document collections and retrieval-augmented generation
- Privacy, reproducibility, and reporting standards

Core question

When can LLMs become part of a research design, not only a personal productivity tool?